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CSCI 2824 Section 001
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1. A. $A=B$
B. A is a subset of B.
C. B is an empty set.
D. Nothing (this statement is always true).
E. $A=B$
2. $(B-A) \cup (C-A) = (B \cup C) \cap (\text{not})A$ by definition of intersection
 $(B \cup C) \cap (\text{not})A = (B \cup C) - A$ by definition of intersection
3. $f(g(x)) = x^2 + 4x + 5$ $g(f(x)) = x^2 + 3$
4. A. The set of all CSCI2824 students to all integers.
B. It's not possible because not everyone is a CSCI2824 student.
5. A. Not bijective because there are no values of m or n for which $f(m,n) < 0$.
B. Bijective.
6. $a_n = (n+1)^2$
7. A. If $a_n = 2-n$, then $a_{n-1} = 3-n$ and $a_{n-2} = 4-n$
 $2-n = 3-n + 2(4-n) + 2n-9$
 $2-n = 2-n$
Q.E.D.
B. If $a_n = 2-n + b \cdot 2^n$, then $a_{n-1} = 3-n + (b/2)2^n$ and $a_{n-2} = 4-n + (b/4)2^n$
 $2-n + b \cdot 2^n = 3-n + (b/2)2^n + 2(4-n + (b/4)2^n) + 2n-9$
 $2-n + b \cdot 2^n = 2-n + b \cdot 2^n$
Q.E.D.
8. C
9. n^2
10. If $R \sim Q$ between 3 and 4 was countable, R would have to be countable because Q is countable. R isn't countable, so this is a contradiction.

